MATH 110 — Week 4

Notes Chapters 7–8; Homework 4

§1 Isomorphisms and inverses

The question. When can a linear map be reversed? The map $T(x, y) = (x + y, y)$ has inverse $S(a, b) = (a - b, b)$, since both compositions return the input.

Definition (Notes Definition 21; LADR 3.59, 3.69)**.** A linear map $T : V \to W$ is invertible, or an isomorphism, if a linear map $S : W \to V$ satisfies $ST = I_V$ and $TS = I_W$, where $I_V(v) = v$ and $I_W(w) = w$. Such an S is an inverse. The spaces are isomorphic, written $V \cong W$, if an isomorphism exists. A left inverse satisfies $ST = I_V$; a right inverse satisfies $TS = I_W$.

Tools used below (Week 3 Propositions 21, 23–26)

1. For a basis $v_1, \dots, v_n$ and arbitrary vectors $w_1, \dots, w_n$, the unique linear map with $Tv_j = w_j$ is $T(\sum a_j v_j) = \sum a_j w_j$. Unique basis coefficients make this a function and verify linearity.
2. T is injective iff $\ker T = \{0\}$: $Tu = Tv$ is equivalent to $T(u - v) = 0$. It is onto iff $\operatorname{range} T$ is the full codomain, by the definition of range.
3. In finite-dimensional V, $\dim V = \dim \ker T + \dim \operatorname{range} T$. Extend a kernel basis u_i by vectors v_j to a basis of V; the Tv_j span the range. If $\sum a_j Tv_j = 0$, then $\sum a_j v_j \in \operatorname{span}(u_i)$, so independence of the full domain basis forces all $a_j = 0$. Thus the Tv_j are a range basis.

Proposition (Notes Proposition 28; LADR 3.60)

An inverse is unique, denoted T^{-1}. More strongly, any left inverse L and right inverse R of the same map satisfy $L = R$.

Where the proof comes from

Want. Show that a left inverse L and a right inverse R agree.

Key step. The two identities contain the same map T: $LT = I_V$ and $TR = I_W$. In $L = LI_W$, replacing I_W by TR puts LT next to R, so both inverse identities can be used.

Proof. 1. $L = LI_W = L(TR) = (LT)R = I_V R = R$.

2. Any two two-sided inverses are in particular a left and right inverse, so they agree.

□

Example (Notes Chapter 7: one-sided nonuniqueness)

Let $A : F^2 \to F$ have row matrix $(1 \;\; 2)$.

1. For every $t \in F$, the column $R_t = (1 - 2t, t)^{\mathsf{T}}$ gives $AR_t = 1$. Different t give different right inverses.
2. There is no left inverse: $A(-2, 1) = 0$ for a nonzero vector, so $LA = I$ would imply $(-2, 1) = L0 = 0$.
3. There are infinitely many right inverses when F is infinite, but only $|F|$ of this displayed family when F is finite.

Proposition (Notes Proposition 29; LADR 3.63)

A linear map is invertible iff it is both injective and surjective. The inverse of a bijective linear map is automatically linear.

Where the proof comes from

Want. Prove the set inverse S respects linear combinations.

Know. $T(Sw) = w$, and T is injective.

Key step. Both $S(aw_1 + w_2)$ and $aSw_1 + Sw_2$ map to $aw_1 + w_2$. Injectivity makes these two inputs equal, without assuming S is linear.

Proof. 1. If T^{-1} exists and $Tv = 0$, then $v = T^{-1}Tv = T^{-1}0 = 0$, so T is injective. Every $w = T(T^{-1}w)$ is in its range, so it is onto.

2. Conversely, define $S(w)$ as the unique v with $Tv = w$. Then $TS = I_W$ and $ST = I_V$.
3. For $a \in F$, $T(S(aw_1 + w_2)) = aw_1 + w_2 = T(aSw_1 + Sw_2)$. Injectivity yields $S(aw_1 + w_2) = aSw_1 + Sw_2$, proving linearity.

□

Proposition (Notes Proposition 30; LADR 3.70)

Two finite-dimensional vector spaces over the same field are isomorphic iff their dimensions agree.

Where the proof comes from

Want. Build an isomorphism from the equality of dimensions.

Key step. Equally long bases allow the rule $T(\sum a_j v_j) = \sum a_j w_j$. Unique coefficients make it linear; sending each w_j back to v_j gives the two inverse identities. In the other direction, rank–nullity with zero kernel gives equal dimensions.

Proof. 1. If $T : V \to W$ is an isomorphism, its kernel is zero and its range is W. Rank–nullity gives $\dim V = \dim W$.

2. If both dimensions are n, choose bases $v_1, \ldots, v_n$ and $w_1, \ldots, w_n$. Define linear maps by $Tv_j = w_j$ and $Sw_j = v_j$.
3. The compositions agree with the identity on their bases and therefore everywhere. Hence T is an isomorphism.

□

Proposition (Notes Chapter 7: infinite-dimensional variant)

Two spaces with bases indexed by the same set are isomorphic. Conversely, an isomorphism sends any basis to a basis.

Where the proof comes from

Want. Use the same construction for bases of any cardinality.

Key step. Even with infinitely many basis vectors, each input uses only finitely many coefficients. Pairing the basis vectors therefore still defines an ordinary finite sum for each input; reversing that pairing gives an inverse.

Proof. 1. Pair the basis vectors by their common index set and define T on finite linear combinations using the same coefficients. Unique basis expansion makes it well-defined and linear; reversing the pairing gives an inverse.

2. Conversely, if T is an isomorphism, its images of a basis span by surjectivity. A finite relation among these images becomes a relation in the original basis after applying T^{-1}, so the images are independent.
3. The existence of arbitrary bases and their cardinal comparison use the choice results cited in Week 2. The notation $\dim V = \infty$ alone does not specify the cardinality of a basis.

□

Proposition (Notes Proposition 31; LADR 3.65)
For $\dim V = \dim W < \infty$ and linear $T : V \to W$, invertibility, injectivity, and surjectivity are equivalent.

Where the proof comes from

Want. Deduce bijectivity from just injectivity or surjectivity.
Know. For the common finite dimension n, rank–nullity says $n = \dim \ker T + \dim \operatorname{range} T$.
Key step. Zero kernel forces range dimension n; full range forces kernel dimension zero. Thus either condition gives both, and Proposition 29 gives the inverse.

Proof. 1. Let $\dim V = \dim W = n$. If T is injective, rank–nullity gives $\dim \operatorname{range} T = n$, so $\operatorname{range} T = W$ by the full-dimension subspace criterion (Week 2 Proposition 14). If T is onto, it gives $\dim \ker T = n - n = 0$, so the kernel is zero. Either condition thus gives bijectivity and Proposition 29 gives invertibility.
2. Invertibility gives both properties by Proposition 29, proving all implications.

□

Compute an inverse by solving for the input

1. If $(a, b) = T(x, y) = (x + y, y)$, solve $y = b$ and $x = a - b$.
2. Thus $T^{-1}(a, b) = (a - b, b)$; substitution verifies $T^{-1}T = I$ and $TT^{-1} = I$.

Shortcuts

- For equal finite dimensions, prove zero kernel or full range; Proposition 31 supplies invertibility.
- To prove two linear maps equal, compare them only on a basis (Week 3).

Traps

- A one-sided inverse need not be unique or two-sided for unequal dimensions: the row $(1 \;\; 2)$ above has multiple right inverses and no left inverse.
- Two abstractly isomorphic spaces do not make every map between them invertible: the zero map on $\mathbb{R}$ is a counterexample.

§2 The two-by-two formula and polynomial existence

The question. For $A = \left(\begin{smallmatrix} 2 & 1 \\ 5 & 3 \end{smallmatrix}\right)$, how can we solve $A(x, y)^{\mathsf{T}} = (r, s)^{\mathsf{T}}$ for arbitrary (r, s)? The combination $2 \cdot 3 - 1 \cdot 5 = 1$ will allow division in the inverse formula.

Definition (LADR 3.79–3.80)**.** A square matrix $A \in F^{n,n}$ is invertible if there is $B \in F^{n,n}$ with $AB = BA = I_n$; its unique inverse is denoted A^{-1}. This is exactly invertibility of $x \mapsto Ax$: for compatible matrices, coordinate i of $A(Bx)$ is $\sum_j a_{ij} \sum_k b_{jk} x_k = \sum_k (\sum_j a_{ij} b_{jk}) x_k$, coordinate i of $(AB)x$. Also $I_n x = x$.

Definition (Notes Chapter 7: two-by-two determinant)**.** For $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ define $\det A = ad - bc$. This notation here is only the two-by-two formula; no general determinant theory is assumed.

Proposition (Notes Chapter 7; HW4 P1)
A is invertible iff $ad - bc \neq 0$. In that case $A^{-1} = \dfrac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.

Where the proof comes from

Want. Solve $ax + by = r$, $cx + dy = s$ for every (r, s).

Key step. Multiply the first equation by d and the second by b and subtract: $(ad - bc)x = dr - bs$. Eliminating x instead gives $(ad - bc)y = as - cr$. These coefficients select the proposed inverse; division requires $ad - bc \neq 0$.

Proof. 1. For $B = \left(\begin{smallmatrix} d & -b \\ -c & a \end{smallmatrix}\right)$, direct multiplication gives $AB = BA = (ad - bc)I_2$: the off-diagonal entries cancel, and each diagonal entry is $ad - bc$.

2. If $ad - bc \neq 0$, scaling B by its reciprocal gives both inverse identities.
3. If $ad - bc = 0$ and $(a, b) \neq (0, 0)$, the nonzero vector $(b, -a)$ satisfies $A(b, -a) = (0, cb - da) = 0$. Thus A is not injective.
4. If $a = b = 0$, the first coordinate of every output is zero, so A is not onto. Proposition 29 excludes invertibility in either case.

□

Proposition (Polynomial root bound used in Notes Example 34)

A nonzero real polynomial of degree at most n has at most n distinct real roots.

Where the proof comes from

Want. Bound the number of distinct roots by the degree.

Know. $x^k - a^k = (x - a)\sum_{j=0}^{k-1} x^{k-1-j}a^j$ for $k \geq 1$.

Key step. A root a gives $p = (x - a)q$. If $b \neq a$ is another root, $0 = (b - a)q(b)$ forces $q(b) = 0$. Thus each distinct root reduces the degree by one, so there cannot be more roots than the original degree.

Proof. 1. If $p(a) = 0$, write $p(x) = p(x) - p(a)$ and factor each monomial difference using the displayed identity. Thus $p = (x - a)q$, with degree reduced by one for nonzero p.

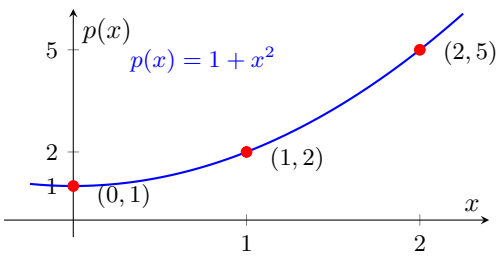
2. If $b \neq a$ is another root, $0 = p(b) = (b - a)q(b)$ implies $q(b) = 0$. Repeat for distinct roots.
3. After r distinct roots have been factored, the degree is at least r. Hence $r \leq n$.

□

Example (Notes Example 34: interpolation)

Let $x_0, \ldots, x_n$ be distinct real numbers and $y_0, \ldots, y_n$ arbitrary.

1. Define $T : \mathcal{P}_n(\mathbb{R}) \to \mathbb{R}^{n+1}$ by $Tp = (p(x_0), \ldots, p(x_n))$. Evaluation is linear.
2. If $Tp = 0$, the root bound forces $p = 0$. Both spaces have dimension $n + 1$, so Proposition 31 gives bijectivity.
3. Thus a unique polynomial satisfies all $p(x_j) = y_j$. A computing formula is $p(x) = \sum_{j=0}^{n} y_j \prod_{k \neq j}(x - x_k)/(x_j - x_k)$: at x_i, every summand is zero except $j = i$, which equals y_i.
4. For points $(0, 1), (1, 2), (2, 5)$, the formula gives $\frac{1}{2}(x - 1)(x - 2) - 2x(x - 2) + \frac{5}{2}x(x - 1) = 1 + x^2$.



Three values at distinct inputs determine one polynomial of degree at most 2. The computed polynomial passes through all three specified points.

Example (Notes Example 35; LADR 3.67)

Given any $q \in \mathcal{P}(\mathbb{R})$, there exists a polynomial p with $((x^2+5x+7)p)'' = q$.

1. Choose $m \geq 0$ with $q \in \mathcal{P}_m(\mathbb{R})$ and define $T(p) = ((x^2+5x+7)p)''$ on $\mathcal{P}_m(\mathbb{R})$. This is linear; a nonzero degree-r input produces degree r output by the power rule, so the map stays in this space.
2. If $Tp = 0$, the product $(x^2+5x+7)p$ has second derivative zero, hence degree at most 1 or is zero. A nonzero p would give degree at least 2, impossible. Thus $p = 0$.
3. Proposition 31 gives surjectivity, proving existence. For example $q = 2$ has solution $p = 1$ by differentiating x^2+5x+7 twice.

Example (Notes Example 36)

On the infinite-dimensional space $\mathcal{P}(\mathbb{R})$, differentiation is onto but not injective.

1. Every $q = \sum_{j=0}^{m} a_j x^j$ is the derivative of $\sum_{j=0}^{m} a_j x^{j+1}/(j+1)$.
2. Nonzero constants have derivative zero, so the kernel is nonzero. Thus the equal finite-dimension criterion cannot be applied here.

Shortcuts

- Two-by-two invertibility needs only $ad - bc$; over a finite field reduce all arithmetic in that field.
- An equal-dimensional polynomial map may be easier to prove injective by degree than to solve coefficient by coefficient.
- For interpolation, use the product formula to compute, and the root bound to prove uniqueness.

Traps

- Distinct interpolation inputs are necessary for arbitrary outputs: $p(0) = 1$ and $p(0) = 2$ cannot both hold.
- The two-by-two inverse requires division by a nonzero determinant: $\left(\begin{smallmatrix} 1 & 2 \\ 2 & 4 \end{smallmatrix}\right)$ has nonzero entries but is not invertible.

§3 Products and one-sided inverses

The question. For $S(x,y) = (x+y, y)$ and $T(x,y) = (2x, y)$ over $\mathbb{R}$, $ST(x,y) = (2x+y, y)$. Which inverse action must be applied first to recover (x,y)?

Proposition (Notes Proposition 32; LADR 3D.2, 3D.11)

If operators S, T are invertible, $(ST)^{-1} = T^{-1}S^{-1}$. If ST is invertible, then S is surjective and T is injective. In finite dimension the latter condition also makes both S, T, and therefore TS, invertible.

Where the proof comes from

Want. Reverse a composition and test what its invertibility forces.

Key step. In $T^{-1}S^{-1}ST$, the adjacent $S^{-1}S$ cancels first, then $T^{-1}T$. For necessity, $Tv = 0$ implies $STv = 0$, so T must be injective. Every target of the onto map ST is also an output of S, so S must be onto.

Proof. 1. $(T^{-1}S^{-1})(ST) = T^{-1}(S^{-1}S)T = I$ and $(ST)(T^{-1}S^{-1}) = S(TT^{-1})S^{-1} = I$. This proves the inverse formula.

2. If ST is invertible, any $v = ST((ST)^{-1}v)$ is an image under S, so S is onto.
3. If $Tv = 0$, then $STv = 0$; injectivity of ST gives $v = 0$. Thus T is injective.
4. In finite dimension Proposition 31 makes both factors invertible, and the first part also applies to TS.

□

Proposition (Notes Chapter 7: restricted factors)

Even without finite dimension, if ST is invertible, then $T : V \to \operatorname{range} T$ and $S|_{\operatorname{range} T} : \operatorname{range} T \to V$ are isomorphisms, and $(ST)^{-1} = T^{-1}(S|_{\operatorname{range} T})^{-1}$, with these restricted domains understood.

Where the proof comes from

Want. Invert the restricted factors when T may fail to be onto V and S may fail to be injective on V.

Key step. T is injective and becomes onto when its target is $\operatorname{range} T$. On that intermediate space, ST being onto forces $S|_{\operatorname{range} T}$ onto, and $S(Tu) = 0$ forces $u = 0$ and hence $Tu = 0$. Thus both restricted factors are bijective.

Proof. 1. The map $T : V \to \operatorname{range} T$ is onto by definition and injective by Proposition 32, hence invertible by Proposition 29.

2. The restriction of S is onto because every $v = STu$ has a preimage $Tu \in \operatorname{range} T$. If $w = Tu$ lies in its kernel, then $STu = 0$, so $u = 0$ and $w = 0$. Thus it too is invertible.
3. The same two-composition calculation as Proposition 32, with these domains, yields the displayed formula.

□

Example (Notes Example 37)

On $V = \mathcal{P}(\mathbb{R})$, let $Sp = p'$ and $(Tp)(x) = \int_0^{x+1} p(t)\,dt$.

1. Termwise integration and differentiation give $STp(x) = p(x+1)$; its inverse sends $p(x)$ to $p(x-1)$.
2. Every Tp vanishes at $x = -1$. Conversely, if $q(-1) = 0$, set $p(x) = q'(x-1)$; then $Tp(x) = q(x) - q(-1) = q(x)$. Thus $\operatorname{range} T = U = \{q : q(-1) = 0\}$.
3. The inverse of $S|_U$ sends p to $\int_{-1}^{x} p(t)\,dt$; the inverse of $T : V \to U$ sends q to $q'(x-1)$. Composing these gives $p(x-1)$ as required.
4. TS sends every constant to zero, so it is not invertible, even though ST is invertible.

Proposition (Notes Proposition 33; LADR 3D.11)

For operators S, T on a finite-dimensional space, $ST = I$ iff $TS = I$.

Where the proof comes from

Want. Obtain $TS = I$ from $ST = I$ in finite dimension.

Know. $ST = I$ forces T injective, so Proposition 31 makes T invertible.

Key step. Multiply $ST = I$ on the right by T^{-1} to get $S = T^{-1}$. This identifies S with the two-sided inverse and supplies $TS = I$.

Proof. 1. If $ST = I$, Proposition 32 gives invertibility of T, so $S = S(TT^{-1}) = (ST)T^{-1} = T^{-1}$. Hence $TS = I$.

2. Interchange S, T to prove the converse.

□

Example (Notes Example 38)

On $\mathcal{P}(\mathbb{R})$, let $Sp = p'$ and $Tp(x) = \int_0^x p(t)\,dt$.

1. Termwise calculus gives $STp = p$, so $ST = I$.
2. It also gives $TSp = p - p(0)$; in particular $TS(1) = 0 \neq 1$. Thus $TS \neq I$ in this infinite-dimensional example.

Shortcuts

- For square finite-dimensional matrices, one inverse product equal to I suffices by Proposition 33.
- When a product is invertible, its first-applied factor must be injective and its last-applied factor must be onto.

Traps

- Invertibility of ST does not force invertibility of TS in infinite dimension: Example 37 is a counterexample.
- Inverses reverse order. For $S = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, $T = \left(\begin{smallmatrix} 2 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ over $\mathbb{R}$, $S^{-1}T^{-1} = \left(\begin{smallmatrix} 1/2 & -1 \\ 0 & 1 \end{smallmatrix}\right)$ while $(ST)^{-1} = \left(\begin{smallmatrix} 1/2 & -1/2 \\ 0 & 1 \end{smallmatrix}\right)$.

§4 Inverse matrices and change of basis

The question. For $T(x, y) = (x, 0)$, compare the standard basis with $B = ((4, 2), (5, 3))$. The output $T(4, 2) = (4, 0)$ has B-coordinates $(6, -4)$ because $6(4, 2) - 4(5, 3) = (4, 0)$. How do the two matrices relate?

Coordinate tools (LADR 3.71–3.78, 3.81)

1. For an ordered basis $B = (b_1, \ldots, b_n)$, write $v = \sum_j a_j b_j$ and define $[v]_B = (a_1, \ldots, a_n)^{\mathsf{T}}$. Unique expansions make the coordinate map linear and bijective. For instance (LADR 3.74), $[2-7x+5x^3+x^4]_{(1,x,x^2,x^3,x^4)} = (2, -7, 0, 5, 1)^{\mathsf{T}}$.
2. For a basis $C = (c_1, \ldots, c_m)$ of W, column j of $M_C^B(T)$ is $[Tb_j]_C$. Expanding $Tv = \sum_j a_j Tb_j$ gives $[Tv]_C = M_C^B(T)[v]_B$.
3. Arbitrary columns prescribe arbitrary basis images, hence exactly one T by the earlier basis rule. Columnwise sums/scaling agree with sums/scaling of maps. Thus $T \mapsto M_C^B(T)$ is an isomorphism $L(V, W) \to F^{m,n}$. The matrices E_{ij} (one 1 at entry (i, j), zeros elsewhere) form a basis of length mn: each $A = \sum_{ij} a_{ij} E_{ij}$, and a zero sum forces each coefficient to be zero, so $\dim L(V, W) = mn$.
4. The Tb_j span range T, and the coordinate isomorphism sends this spanning list to the columns. Thus $\dim \operatorname{range} T = \operatorname{rank} M_C^B(T)$.
5. For $T : U \to V$, $S : V \to W$, and bases B, C, D, applying the coordinate formula to each STb_j gives $M_D^B(ST) = M_D^C(S)M_C^B(T)$. The shared middle basis is C. The identity map in a single basis has matrix I, since its columns are the standard coordinate columns.

Definition (LADR 3.82)**.** For bases B, C of the same finite-dimensional space, the transition matrix $Q = M_C^B(I)$ has columns $[b_j]_C$ and satisfies $[v]_C = Q[v]_B$.

Proposition (LADR 3.82, 3.86: transition and inverse matrices)
$Q^{-1} = M_B^C(I)$. If $T : V \to W$ is invertible, B is a basis of V and C a basis of W, then $M_B^C(T^{-1}) = (M_C^B(T))^{-1}$.

Where the proof comes from

Want. Reverse a coordinate transition or an invertible map.

Key step. Write matrices of both identity compositions. The middle bases agree in each product, so the composition rule gives the two matrix inverse equations.

Proof. 1. The identity compositions give $M_B^C(I)M_C^B(I) = I_n$ and $M_C^B(I)M_B^C(I) = I_n$. These are precisely the inverse equations for Q.

2. Likewise, $T^{-1}T = I_V$ gives $M_B^C(T^{-1})M_C^B(T) = I_n$, and $TT^{-1} = I_W$ gives $M_C^B(T)M_B^C(T^{-1}) = I_m$. Here $m = n$ because T is an isomorphism. Thus the matrices are inverses.

□

Proposition (LADR 3.84: change of basis)
For $T \in L(V)$, let $A = M_B^B(T)$, $D = M_C^C(T)$, and $Q = M_C^B(I)$. Then $QA = DQ$ and $A = Q^{-1}DQ$.

Where the proof comes from

Want. Compare the two matrices of the same operator.
Key step. Compute $[Tv]_C$ in two ways: apply T in B then change coordinates, or change coordinates first then apply T in C. The two routes give $QA[v]_B$ and $DQ[v]_B$, so they must agree for every coordinate column.

Proof. 1. By the transition and coordinate formulas, $[Tv]_C = Q[Tv]_B = QA[v]_B$ and $[Tv]_C = D[v]_C = DQ[v]_B$.
2. Every standard column occurs as $[v]_B$ (take $v = b_j$). Thus corresponding columns of QA, DQ agree, giving $QA = DQ$. Multiplying on the left by Q^{-1} gives the formula.

□

Proposition (LADR 3.68, 3.80: inverse identities)
If $\dim V = \dim W < \infty$, $T : V \to W$ and $S : W \to V$ are linear, then $ST = I_V$ iff $TS = I_W$. Also $(A^{-1})^{-1} = A$, and $(AB)^{-1} = B^{-1}A^{-1}$ for invertible square matrices.

Where the proof comes from

Want. Check one inverse equation between different spaces of the same finite dimension.
Key step. $ST = I_V$ implies $Tv = 0 \Rightarrow v = STv = 0$. Equal dimensions then give T^{-1}, and $S = S(TT^{-1}) = (ST)T^{-1} = T^{-1}$. The inverse identities now give the other product and the matrix formulas.

Proof. 1. If $ST = I_V$ and $Tv = 0$, then $v = STv = 0$. Proposition 31 makes T invertible, and $S = S(TT^{-1}) = (ST)T^{-1} = T^{-1}$. Thus $TS = I_W$; interchanging the maps gives the converse.
2. The two equations defining A^{-1} also say that A is its inverse, giving $(A^{-1})^{-1} = A$. Proposition 32 applied to coordinate maps gives the reverse-order product formula.

□

Example (LADR 3.62–3.64: contrasting maps) 1. On $\mathbb{R}^3$, $T(x, y, z) = (-y, x, 4z)$ has inverse $(a, b, c) \mapsto (b, -a, c/4)$, verified by both substitutions.
2. On $\mathcal{P}(\mathbb{R})$, multiplication by x^2 sends $p = \sum_{j=0}^m a_j x^j$ to $\sum_{j=0}^m a_j x^{j+2}$. A zero output forces every $a_j = 0$, so it is injective. Every output has zero constant coefficient; hence 1 is not in its range and it is not onto.
3. On infinite sequences, the backward shift $(a_0, a_1, \ldots) \mapsto (a_1, a_2, \ldots)$ is onto (prepend any coordinate), but not injective (the first coordinate is unrestricted). These show why finite dimension matters.

Example (LADR 3.83: coordinate transitions)
In $\mathbb{R}^2$, let $B = ((4, 2), (5, 3))$ and let C be the standard basis.

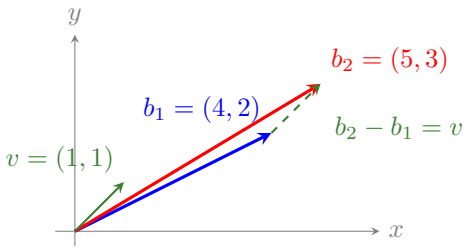
1. The transition $Q = M_C^B(I) = \begin{pmatrix} 4 & 5 \\ 2 & 3 \end{pmatrix}$ has determinant 2, hence the two columns form a basis, with $Q^{-1} = \begin{pmatrix} 3/2 & -5/2 \\ -1 & 2 \end{pmatrix}$.
2. For $T(x, y) = (x, 0)$, $D = M_C^C(T) = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$. Thus $M_B^B(T) = Q^{-1}DQ = \begin{pmatrix} 6 & 15/2 \\ -4 & -5 \end{pmatrix}$.
3. Directly, $T(4, 2) = (4, 0) = 6(4, 2) - 4(5, 3)$ and $T(5, 3) = (5, 0) = \frac{15}{2}(4, 2) - 5(5, 3)$, verifying the columns.

Shortcuts

- Label the direction of every coordinate transition: $Q = M_C^B(I)$ sends B-coordinates to C-coordinates.
- To compute in a new basis, use $QA = DQ$ first; solving it for A determines the order of the factors.

Traps

- Changing only the domain basis gives $DQ = \left(\begin{smallmatrix} 4 & 5 \\ 0 & 0 \end{smallmatrix}\right)$ in the example; changing both gives $Q^{-1}DQ = \left(\begin{smallmatrix} 6 & 15/2 \\ -4 & -5 \end{smallmatrix}\right)$. The two operations are different.
- A matrix inverse has reversed domain/codomain bases: its columns express preimages in the original domain basis.



$$\underbrace{\begin{pmatrix} 4 & 5 \\ 2 & 3 \end{pmatrix}}_{Q} \underbrace{\begin{pmatrix} -1 \\ 1 \end{pmatrix}}_{[v]_B} = \underbrace{\begin{pmatrix} 1 \\ 1 \end{pmatrix}}_{[v]_C}$$

The columns of Q are the B-basis vectors written in the standard basis C. The same vector $v = b_2 - b_1$ has coordinates $(-1, 1)$ in B and $(1, 1)$ in C.

§5 Products and external direct sums

The question. In $\mathcal{P}_1(\mathbb{R}) \times \mathbb{R}^2$, adding $(1 + x, (2, 3))$ and $(x, (1, -1))$ gives $(1 + 2x, (3, 2))$. How do bases and dimensions combine under these componentwise operations?

Definition (Notes Chapter 8; LADR 3.87)**.** The Cartesian product $V_1 \times \cdots \times V_m$ consists of ordered lists $(v_1, \ldots, v_m)$ with $v_j \in V_j$. All spaces use the same field. Addition and scalar multiplication act componentwise.

Example (LADR 3.88: product arithmetic)

In $\mathcal{P}_5(\mathbb{R}) \times \mathbb{R}^3$, let $v = (5 - 6x + 4x^2, (3, 8, 7))$ and $w = (x + 9x^5, (2, 2, 2))$. Adding the polynomial coefficients and vector coordinates gives

$$v + w = (5 - 5x + 4x^2 + 9x^5, (5, 10, 9)), \qquad 2v = (10 - 12x + 8x^2, (6, 16, 14)).$$

The polynomial and the triple stay in their respective components.

Proposition (Notes Proposition 34; LADR 3.89)

The componentwise operations make $V_1 \times \cdots \times V_m$ a vector space. Its zero is $(0, \ldots, 0)$ and the negative of $(v_1, \ldots, v_m)$ is $(-v_1, \ldots, -v_m)$.

Where the proof comes from

Want. Verify vector-space identities for tuples.

Key step. Each coordinate of an asserted identity is the corresponding identity in its factor space. For instance, coordinate j of $a(v + w)$ is $a(v_j + w_j) = av_j + aw_j$, exactly coordinate j of $av + aw$. Zero and negatives are determined in the same way.

Proof. 1. Componentwise sums and scalar multiples remain in the product by closure in each V_j. Associativity and commutativity of addition hold in each coordinate.

2. The displayed zero and negative satisfy their identities in every coordinate. The scalar identity, scalar associativity, and both distributive laws hold coordinatewise by the corresponding axiom in V_j.

□

Example (Notes Examples 39–40) 1. $\mathbb{R}^n$ is the product of n copies of $\mathbb{R}$: the tuples and componentwise operations coincide.
2. The concatenation map $\mathbb{R}^r \times \mathbb{R}^s \to \mathbb{R}^{r+s}$ sends $((x_1,\dots,x_r),(y_1,\dots,y_s))$ to $(x_1,\dots,x_r,y_1,\dots,y_s)$. It is linear; splitting after coordinate r is its linear inverse.
3. Rebracketing $(v,(w,z)) \leftrightarrow ((v,w),z) \leftrightarrow (v,w,z)$ and swapping $(v,w) \leftrightarrow (w,v)$ are likewise linear bijections, verified componentwise.

Definition (Notes Chapter 8: coordinate subspaces)**.** Write $V_{(j)}$ for the subspace of the product with all coordinates zero except possibly coordinate j. The insertion map $v \mapsto (0,\dots,v,\dots,0)$ is linear and bijective onto $V_{(j)}$, with inverse reading coordinate j; thus $V_j \cong V_{(j)}$.

Proposition (Notes Proposition 35)

$V_1 \times \cdots \times V_m = V_{(1)} \oplus \cdots \oplus V_{(m)}$. For this reason the product is called an external direct sum; its factors need not start as subspaces of one common space.

Where the proof comes from

Want. Give a unique decomposition into coordinate subspaces.

Key step. In $(v_1,\dots,v_m)$, only the $V_{(j)}$ summand can contribute to coordinate j. That coordinate must be v_j, so inserting each coordinate gives existence and also forces every summand.

Proof. 1. Every tuple equals $(v_1,0,\dots,0)+\cdots+(0,\dots,0,v_m)$, so the subspaces span the product.
2. In any such decomposition, coordinate j of the total equals coordinate j of its $V_{(j)}$ summand. Thus each summand is forced; the sum is direct.

□

Proposition (Notes Proposition 36; LADR 3.92)

If all V_j are finite-dimensional, then $\dim(V_1 \times \cdots \times V_m) = \sum_{j=1}^{m} \dim V_j$.

Where the proof comes from

Want. Count a basis of the product.

Key step. Insert a basis of V_j into coordinate j. Any tuple expands using these inserted vectors; a zero relation gives a zero relation in each factor basis, so all coefficients vanish. The resulting basis has $\sum_j \dim V_j$ vectors.

Proof. 1. Inserting a finite basis of V_j into coordinate j gives a basis of $V_{(j)}$: spanning and independence hold coordinatewise. Thus $V_{(j)}$ is finite-dimensional with dimension $\dim V_j$.
2. Apply the direct-sum dimension formula to Proposition 35 and add these dimensions.

□

Proposition (Notes Proposition 37; LADR 3.93–3.94)

For subspaces $U_1,\dots,U_m \subseteq V$, the addition map $\Gamma : \prod_j U_j \to \sum_j U_j$, $\Gamma(u_1,\dots,u_m) = \sum_j u_j$, is linear and surjective. It is injective iff the sum is direct. If the U_j are finite-dimensional, $\dim(\sum_j U_j) \le \sum_j \dim U_j$, with equality iff the sum is direct.

Where the proof comes from

Want. Determine when adding tuple components loses uniqueness.

Key step. Two tuples have the same sum exactly when their difference satisfies $\sum_j u_j = 0$. Hence $\ker \Gamma = \{(u_j) : \sum_j u_j = 0\}$, which is zero exactly for a direct sum. Rank–nullity then measures the dimension difference by $\dim \ker \Gamma$.

Proof. 1. Addition commutes with linear combinations, so Γ is linear. Every element of the sum is by definition some $\sum u_j$, proving surjectivity.
2. The kernel consists of tuples with $\sum u_j = 0$. It is zero iff every such tuple is zero, exactly the direct-sum criterion.
3. In finite dimension, rank–nullity gives $\dim(\sum U_j) = \sum \dim U_j - \dim \ker \Gamma$. The last term is nonnegative and is zero iff Γ is injective.

□

Construct a product basis

1. A basis $v_1, \dots, v_r$ of V and $w_1, \dots, w_s$ of W give $(v_1, 0), \dots, (v_r, 0), (0, w_1), \dots, (0, w_s)$ in $V \times W$.
2. Any (v, w) expands using the two coordinate expansions, proving spanning. A zero relation separates into a zero relation in each basis, so every coefficient is zero.
3. For $\mathbb{R}^2 \times \mathbb{R}^3$, the resulting five vectors have one coordinate 1 and all others zero, grouped into a pair and a triple.
4. For $\mathcal{P}_2(\mathbb{R}) \times \mathbb{R}^2$ (LADR 3.91), the basis is $(1, 0), (x, 0), (x^2, 0), (0, (1, 0)), (0, (0, 1))$, where the first three zeros denote the zero vector of $\mathbb{R}^2$.

$$(1 + 2x, (3, 4)) = \underbrace{(1, (0, 0)) + 2(x, (0, 0))}_{\text{polynomial-coordinate subspace}} + \underbrace{3(0, (1, 0)) + 4(0, (0, 1))}_{\text{vector-coordinate subspace}}.$$

In $\mathcal{P}_1(\mathbb{R}) \times \mathbb{R}^2$, each component contributes its own basis vectors. Reading the two components separately recovers all four coefficients, so the displayed decomposition is unique.

Shortcuts

- A product dimension is a sum, because component bases contribute separate basis vectors.
- To check whether a sum is direct, test whether its addition map has zero kernel.

Traps

- $\dim(V \times W)$ is not generally $(\dim V)(\dim W)$: $\mathbb{R}^2 \times \mathbb{R}^3$ has dimension 5, not 6.
- A product and an internal sum need not have equal dimension: for $U = W = \mathbb{R}$ inside $\mathbb{R}$, $U \times W$ has dimension 2 but $U + W$ dimension 1.

§6 Affine sets and quotient operations

The question. When do two translated copies of a subspace represent the same set? The lines $(1, 0) + \mathbb{R}(0, 1)$ and $(1, 4) + \mathbb{R}(0, 1)$ are equal because their base points differ vertically.

Definition (Notes Definitions 22–23; LADR 3.95, 3.97)**.** For a subspace $U \subseteq V$ and $v \in V$, the translate $v + U = \{v + u : u \in U\}$ is an affine set parallel to U. Two affine sets are parallel if both are translates of the same subspace. The word coset also means a translate $v + U$.

Example (Notes Examples 41–42) 1. The line $\{(x, y) : x + y = 1\}$ equals $(1, 0) + \operatorname{span}((-1, 1))$, because its points are $(1 - t, t)$ for arbitrary real t.
2. The polynomials with derivative x^2 are $\frac{1}{3}x^3 + \mathbb{R}$, where $\mathbb{R}$ denotes the subspace of constant polynomials. Differentiation checks inclusion; subtracting $\frac{1}{3}x^3$ gives a polynomial $r = \sum_{j=0}^{m} a_j x^j$ with $r' = \sum_{j=1}^{m} j a_j x^{j-1} = 0$. Each $j a_j = 0$, so $a_j = 0$ for $j \geq 1$ over $\mathbb{R}$; only the constant a_0 remains.
3. For fixed $U = \mathbb{R}(0, 1)$, the cosets in $\mathbb{R}^2$ are exactly the vertical lines: $(a, b) + U = \{(a, t) : t \in \mathbb{R}\}$.

Example (LADR 3.96, 3.98: translates of lines and planes) 1. For $U = \{(t, 2t) : t \in \mathbb{R}\}$, $(17, 20) + U = \{(17 + t, 20 + 2t) : t \in \mathbb{R}\}$. Writing $x = 17 + t$ gives $y = 2x - 14$: the line through $(17, 20)$ parallel to U. Every slope-2 line $y = 2x + c$ equals $(0, c) + U$.

2. If $U = \{(x, y, 0) : x, y \in \mathbb{R}\} \subseteq \mathbb{R}^3$, then $(a, b, c) + U = \{(s, t, c) : s, t \in \mathbb{R}\}$, the plane $z = c$. Thus its translates are exactly the planes parallel to the xy-plane.
3. For a line $U = \mathbb{R}u \subseteq \mathbb{R}^3$ with $u \neq 0$, its translates are the lines $v + \mathbb{R}u$ with direction u. In each case the parameter description proves both inclusions.

Proposition (Notes Proposition 38)

An affine set is parallel to exactly one subspace.

Where the proof comes from

Want. Recover the direction subspace from the affine set itself.

Key step. Differences of two points of $v + U$ lie in U, and every $u \in U$ is the difference $(v + u) - v$. Thus the set of point differences determines U. The proof compares two translated descriptions by these same subtractions.

Proof. 1. Suppose $A = v + U = x + W$. Since $v \in x + W$, write $v = x + w_0$ for some $w_0 \in W$.

2. For $u \in U$, the point $v + u$ is in $x + W$, so $v + u = x + w$ for some $w \in W$. Subtract to get $u = w - w_0 \in W$. Thus $U \subseteq W$.
3. Interchange the roles of U, W to obtain the reverse inclusion, hence equality.

□

Proposition (Notes Proposition 39; LADR 3.101)

For a fixed subspace U, the following are equivalent: $v - w \in U$; $v + U = w + U$; $(v + U) \cap (w + U) \neq \varnothing$. Thus two cosets are either identical or disjoint.

Where the proof comes from

Want. Test when two translates agree or intersect.

Key step. A common point $v + u_1 = w + u_2$ gives $v - w = u_2 - u_1 \in U$. Conversely, $v - w \in U$ lets us rewrite $v + u = w + (v - w + u)$ for every $u \in U$, so the whole first translate is in the second.

Proof. 1. If $v - w \in U$, then every $v + u = w + (v - w + u) \in w + U$. Reversing v, w gives equality.

2. If the cosets are equal, they intersect because $v = v + 0$ belongs to $v + U$.
3. If a point equals $v + u_1 = w + u_2$, then $v - w = u_2 - u_1 \in U$. This proves all three implications.

□

Definition (Notes Definition 24; LADR 3.99, 3.102)**.** The quotient V/U is the set of all cosets $v + U$. Define $(v+U)+(w+U) = (v+w)+U$ and $a(v+U) = (av)+U$. These operations must be independent of the representatives v, w, as proved next.

Example (LADR 3.100: elements of a quotient)

For $U = \{(t, 2t) : t \in \mathbb{R}\}$, an element of $\mathbb{R}^2/U$ is a whole slope-2 line. For a line through 0 in $\mathbb{R}^3$, elements of $\mathbb{R}^3/U$ are the parallel lines. For a plane through 0, they are the parallel planes. These follow directly from the descriptions above and the definition of quotient as the set of translates.

Proposition (Notes Definition 24: well-defined operations; LADR 3.103)

The quotient operations are well-defined and make V/U a vector space. Its zero vector is the coset $U = 0 + U$, and the negative of $v + U$ is $-v + U$.

Where the proof comes from

Want. Make addition and scaling independent of the chosen representatives.

Know. Equal cosets mean representative differences lie in U (Proposition 39).

Key step. Changing v, w to v', w' changes their sum by $(v-v')+(w-w') \in U$, and changes av by $a(v-v') \in U$. These are exactly the differences that leave the output coset unchanged.

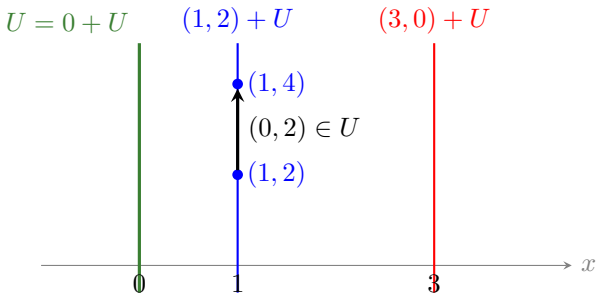
Proof. 1. If $v_1+U = v_2+U$ and $w_1+U = w_2+U$, then $v_1-v_2, w_1-w_2 \in U$. Their sum is in U, so $(v_1+w_1)+U = (v_2+w_2)+U$.

2. Also $a(v_1-v_2) \in U$, so $av_1+U = av_2+U$. Both operations are well-defined.
3. Associativity and commutativity of addition, both distributive laws, scalar associativity, and scalar identity follow by performing the corresponding operations on representatives in V.
4. $(v+U)+(0+U) = v+U$ and $(v+U)+(-v+U) = 0+U$, verifying the asserted zero and negative.

□

Compute with cosets

1. For $U = \mathbb{R}(0,1)$, $(1,2)+U = (1,4)+U$ because the representative difference $(0,-2)$ lies in U.
2. Adding $(1,2)+U$ and $(3,-1)+U$ gives $(4,1)+U = (4,0)+U$. Multiplication by 2 gives $2((1,2)+U) = (2,4)+U = (2,0)+U$.
3. Distinct first coordinates give distinct cosets by Proposition 39. The zero coset is the entire vertical axis U, not merely the point $(0,0)$.



For $U = \mathbb{R}(0,1)$, the two marked representatives give the same blue coset: their difference is in U. The red coset is disjoint. The zero vector of the quotient is the entire green line U.

Shortcuts

- To compare cosets, subtract their representatives and test membership in U.
- To add or scale cosets, choose the simplest representative; well-definedness guarantees the result is independent of that choice.

Traps

- An affine set need not be a subspace: $x+y=1$ excludes zero.
- Parallel affine sets need not intersect: the vertical lines $x=1$ and $x=3$ are disjoint.
- A coset is a set of vectors, not a chosen vector: $(1,2)+U = (1,4)+U$ above although $(1,2) \neq (1,4)$.

§7 Problem set

HW4 P1. For $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, (a) verify $A\left(\begin{smallmatrix} d & -b \\ -c & a \end{smallmatrix}\right) = (ad-bc)I$. (b) Define $\det A = ad-bc$; prove nonzero determinant implies invertibility and derive the inverse. (c) Prove zero determinant implies noninvertibility without general determinant theory.
Hint: Use the two-by-two proposition and its explicit kernel vector.

HW4 P2. Decide invertibility and find inverses when they exist: $\left(\begin{smallmatrix} 2 & 1 \\ 5 & 3 \end{smallmatrix}\right)$ over F, $\left(\begin{smallmatrix} 1 & i \\ i & -1 \end{smallmatrix}\right)$ over $\mathbb{C}$, and $\left(\begin{smallmatrix} 1 & 2 \\ 3 & 1 \end{smallmatrix}\right)$ over $\mathbb{F}_5$.
Hint: Compute $ad - bc$ in the stated field.

HW4 P3 (LADR 3D.3). For $T \in L(V)$ with V finite-dimensional, prove equivalence: (a) T is an isomorphism; (b) for every basis $v_1, \ldots, v_n$ of V, the list $Tv_1, \ldots, Tv_n$ is a basis of V.
Hint: Use injectivity for image independence and surjectivity for image spanning.

HW4 P4. For matrices B, C with BC defined, (a) prove $\operatorname{rank}(BC) \leq \min\{\operatorname{rank} B, \operatorname{rank} C\}$. (b) Give a strict example with both factors of positive rank.
Hint: Use $\operatorname{range}(BC) \subseteq \operatorname{range} B$ and restrict B to $\operatorname{range} C$.

HW4 P5. For finite-dimensional V, W, prove $\dim(V \times W) = \dim V + \dim W$ by constructing a product basis from bases of the factors.
Hint: Use the two coordinate embeddings from Proposition 35.

HW4 P6. For $U \subseteq V$ a subspace, (a) prove $v + U$ is a subspace iff $v \in U$, in which case $v + U = U$. (b) Prove $v + U = v' + U$ iff $v - v' \in U$.
Hint: For (a) test whether zero belongs; for (b) apply Proposition 39.

HW4 P7 (optional; source wording). Suppose U is a subspace of finite-dimensional V, and $S : U \to V$ is injective. Prove S extends to an invertible operator $T : V \to V$. The source omits the necessary assumption that S is linear; as written the assertion is false. Give a counterexample, then prove the intended statement with linearity added.
Hint: Try $S(x) = x^3$ on $\mathbb{R}$; for the linear case extend a basis and its independent image list.

Exam practice (Practice Midterm 1, Question 2). Let V, W be finite-dimensional and $T : V \to W$ injective and linear. (a) For a basis $v_1, \ldots, v_n$ of V, prove $Tv_1, \ldots, Tv_n$ independent. (b) Construct a linear $S : W \to V$ with $ST = I_V$.
Hint: Extend the independent image list to a basis of W and prescribe S on that basis.

Exam practice (practice-midterm1.pdf, Question 3). Define $T : \mathbb{R}^3 \to \mathcal{P}_2(\mathbb{R})$ by $T(a, b, c) = (a + b) + (b + c)x + (a + c)x^2$. Use domain basis $B = ((1, 0, 0), (1, 1, 0), (1, 1, 1))$ and codomain basis $C = (1, x, x^2)$. (a) Find $M_C^B(T)$. (b) Prove T is an isomorphism.
Hint: Compute the three basis images, then test the kernel and apply Proposition 31.

Exam review (Midterm 1 Review, Section II, Exercise 9). Let U, W be subspaces of finite-dimensional V with $U \cap W = \{0\}$. Construct an isomorphism $U \times W \to U \oplus W$.
Hint: Use the addition map of Proposition 37; if $u + w = 0$, then $u = -w$ lies in the intersection.

Exam review (Midterm 1 Review, Section II, Exercise 10). On F^∞, define $R(a_1, a_2, \ldots) = (0, a_1, a_2, \ldots)$. (a) Show R is injective but not onto. (b) Explain why no operator on a finite-dimensional space has this property.
Hint: Read coordinates after the first; test the target $(1, 0, \ldots)$ and then Proposition 31.